**START HERE · read the maths.** $\int_a^b f(x)\,dx$: only $x$ changes, from $a$ to $b$; every other letter is temporarily constant. Integrate to $F(x)$, then calculate $[F(x)]_a^b = F(b) - F(a)$ (**upper minus lower**). In $\int_c^d \left[\int_{a(y)}^{b(y)} f(x,y)\,dx\right]dy$, do the bracketed **inner $x$-integral first**, substitute its limits, then do the **outer $y$-integral**. $\sum_{n=1}^{N}$ = add the $N$ terms · $\prod$ = multiply them · $\bar{x} = \frac{1}{N}\sum x_n$ · $\propto$ = "equal apart from a constant".

**Symbols.** $\theta$ = unknown parameter · $\hat{\theta}$ = estimate · $x[n]$ or $x_1,\dots,x_N$ = data · $E[\cdot]$ = expected/long-run average · $\mathrm{Var}(x) = E[x^2] - E[x]^2$ · $r[k]$ = correlation $k$ steps apart · $P(\theta)$ = power spectral density (PSD) · $\sigma^2$ = noise variance · $Q(t) = P(Z > t)$ for standard Gaussian $Z$.

## PART 1 · Probability (1.5 p)

### 1A · Joint PDF: region, integrals, marginals

**IF YOU SEE $p(x,y)$ with a region, or "find $c$", "probability", "marginal", "conditional", "independent", "domain"**

**Meaning.** $p(x,y)$ describes how likely the pair $(x,y)$ is. The **allowed region** (also called support/domain) is where the given formula applies; outside it, $p = 0$. Integrating $p$ over the WHOLE allowed region must give 1—that is the reason "find $c$" works. **If asked for the probability of a condition:** shade only the part of the allowed region where that condition is also true, then integrate $p$ over the shaded part using the same slice method below. This answer is a probability; **do not set it equal to 1**.

**LIMITS — copy this procedure; do not guess.** For $dx\,dy$, $x$ is INNER and $y$ is OUTER. **(1)** Draw the allowed region and label each boundary equation. **(2)** Fix outer $y$ at one allowed value; draw a horizontal slice. **(3)** Where the slice enters/leaves the region gives the lower/upper $x$-limits; solve the boundary for $x$ if needed. These limits may contain $y$. **(4)** Sweep the slice across the whole region: the smallest/largest possible $y$ are numerical outer limits. For $dy\,dx$, swap every word $x \leftrightarrow y$ and use a vertical slice. Rectangle → constant limits; non-rectangle → at least one limit depends on the outer letter.

**FULL EXAMPLE (Apr25), no skipped steps.** $p(x,y) = cx + 1$ on $x \ge 0$, $y \ge 0$, $x + y < 1$. The boundaries meet at $(0,0)$, $(1,0)$, $(0,1)$; the sloped boundary is $y = 1 - x$, equivalently $x = 1 - y$. For $dx\,dy$, a horizontal slice has $0 \le x \le 1 - y$, and slices cover $0 \le y \le 1$.




**A · Find $c$ (normalise).** Write the whole double integral before calculating: $1 = \int_{y=0}^{1}\left[\int_{x=0}^{1-y}(cx+1)\,dx\right]dy$. **Inner, $x$ changes; $y, c$ are constants:** $\int_0^{1-y}(cx+1)dx = \left[\frac{cx^2}{2} + x\right]_0^{1-y} = \frac{c}{2}(1-y)^2 + (1-y) - 0$. **Outer, only $y$ remains.** Expand $(1-y)^2 = 1 - 2y + y^2$: $1 = \int_0^1 \left[\frac{c}{2}(1 - 2y + y^2) + 1 - y\right]dy = \left[\frac{c}{2}\left(y - y^2 + \frac{y^3}{3}\right) + y - \frac{y^2}{2}\right]_0^1$. Substitute $y = 1$, then subtract the value at $y = 0$ (zero): $1 = \frac{c}{2}(1 - 1 + \frac{1}{3}) + 1 - \frac{1}{2} = \frac{c}{6} + \frac{1}{2}$. Thus $\frac{c}{6} = \frac{1}{2} \Rightarrow \boxed{c = 3}$. **Check:** $c = 3$ makes $p = 3x + 1 \ge 0$ throughout the triangle.

**B · Marginal = integrate away the OTHER variable.** For $p_X(x)$, use a vertical slice: fixed $x$, $0 \le y \le 1 - x$; the region's $x$-shadow is $0 \le x \le 1$: $p_X(x) = \int_0^{1-x}(3x+1)dy = [3xy + y]_0^{1-x} = (3x+1)(1-x) = 1 + 2x - 3x^2$, and $p_X(x) = 0$ otherwise. **Check:** $\int_0^1(1 + 2x - 3x^2)dx = [x + x^2 - x^3]_0^1 = 1$. Similarly $p_Y(y) = \int_0^{1-y}(3x+1)dx = \frac{3}{2}(1-y)^2 + (1-y) = \frac{5}{2} - 4y + \frac{3}{2}y^2$ for $0 \le y \le 1$, 0 otherwise.

**C · Conditional = joint ÷ marginal of the variable GIVEN.** $p(x \mid y) = p(x,y)/p_Y(y)$. Here $p(x \mid y) = \dfrac{3x+1}{\frac{5}{2} - 4y + \frac{3}{2}y^2}$ for $0 \le x \le 1 - y$ (with $0 \le y < 1$); outside, 0. For $p(y \mid x)$, divide by $p_X(x)$ instead. **Do not reuse the original two-variable range without saying which variable is fixed.**

**D · Independent?** Both must hold: **(1)** support is a rectangle (possibly infinite), **(2)** $p(x,y) = p_X(x)p_Y(y)$ everywhere. Here the triangle fails test 1 → **not independent**. Square example: $p = 6x^2y$ on $[0,1]^2$ gives $p_X = 3x^2$, $p_Y = 2y$, whose product is $6x^2y \to$ independent.

**Fast checks:** every pdf/marginal $\ge 0$ and integrates to 1 on its stated domain · every probability lies in $[0,1]$ · conditional integrates to 1 over the remaining variable's slice · polynomial rule $\int x^n dx = x^{n+1}/(n+1)$, $n \ne -1$.

### 1B · WSS / ergodic (multiple choice)

**IF YOU SEE "WSS"/"wide-sense stationary"/"stationary", "ergodic", dice, random phase, random amplitude**

**Definitions. WSS** requires BOTH: mean $m(t) = E[x(t)]$ is constant, and $r(t_1,t_2) = E[x(t_1)x(t_2)]$ depends only on the lag $\tau = t_1 - t_2$. **Ergodic in the mean** = the time average of EACH realisation equals the ensemble mean $E[x(t)]$. Ensemble average = average across all possible signals at fixed $t$; time average = average along one recorded signal.

1. **Mean test:** calculate $E[x(t)]$; deterministic functions of $t$ stay, random letters become their means. If $t$ remains → not WSS; stop.
2. **Correlation test:** multiply $x(t_1)x(t_2)$, expand, then take $E$. Use $\cos a\cos b = \frac{1}{2}[\cos(a+b) + \cos(a-b)]$ and $E[\cos(k+U)] = 0$ for uniform phase $U$. If only $t_1 - t_2$ remains → WSS. If $t_1 + t_2$ or a separate $t$ remains → not WSS.
3. **Ergodic test — actually compare averages; do not use "random letter frozen" as a shortcut.** Compute $\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T} x(t)dt$ (or the discrete sum) for one realisation and compare with step 1. For ergodicity in variance, repeat with $x^2$ and compare $E[x^2]$.

**Random phase,** $x(t) = \cos(\omega t + U)$, $U$ **uniform on** $[0, 2\pi)$**:** $E[x(t)] = 0$, so mean passes. $r = E[\cos(\omega t_1 + U)\cos(\omega t_2 + U)] = \frac{1}{2}\cos(\omega(t_1 - t_2)) + \frac{1}{2}E[\cos(\omega(t_1 + t_2) + 2U)] = \frac{1}{2}\cos(\omega(t_1 - t_2)) \to$ **WSS**. One realisation is still a cosine, whose long-time average is 0 → **ergodic in the mean**, even though its phase is fixed.

**Random amplitude,** $x(t) = A$ **for all** $t$**:** $E[x] = E[A]$ and $r = E[A^2]$ are constants → **WSS**. But one realisation has time average $A$, not generally $E[A] \to$ **NOT ergodic in the mean**. Fresh independent dice/samples at every time → time average sees the full distribution → ergodic under the exam's usual assumptions; streamed values uniform in $[-A, A]$ have mean 0.

**Vocabulary:** random process = set of possible signals; realisation = one recorded signal; random variable = one numerical outcome. Strict stationarity concerns whole distributions; WSS checks only mean and correlation.

### 1C · Spectral factorisation

**IF YOU SEE $P(\theta) = \dfrac{\text{number}}{A + B\cos\theta}$ + "spectral factorisation"/"factorization", "innovation filter $L(z)$", "$\sigma_i^2$"**

1. Read $A$ and $B$ off the denominator $A + B\cos\theta$ **with their signs** ($25 - 24\cos\theta$: $A = 25$, $B = -24$). Two candidate roots $r = \dfrac{A \pm \sqrt{A^2 - B^2}}{B}$: keep the one between $-1$ and 1, call it $a$.
2. $C = \dfrac{B}{2a}$ (just a helper number).
3. $\sigma_i^2 = \dfrac{\text{number on top}}{C}$, $L(z) = \dfrac{1}{1 + a\,z^{-1}}$.
4. Check $|a| < 1$ and $\sigma_i^2 > 0$ — a negative $\sigma_i^2$ means you kept the wrong root. Sanity: $\dfrac{\sigma_i^2}{1 + a^2 + 2a\cos\theta}$ must give back the $P(\theta)$ you started from.

*Apr24,* $P = \frac{7}{25 - 24\cos\theta}$*:*

**Step 1 — root:** $A = 25$, $B = -24$, so $\sqrt{A^2 - B^2} = \sqrt{625 - 576} = 7$. Candidates: $\frac{25+7}{-24} = -\frac{4}{3}$ and $\frac{25-7}{-24} = -\frac{3}{4}$. Keep $a = -\frac{3}{4}$ because it is between $-1$ and 1.

**Step 2 — helper** $C$**:** $C = \frac{B}{2a} = \frac{-24}{2(-3/4)} = \frac{-24}{-3/2} = 16$.

**Step 3 — answers:** $\sigma_i^2 = \frac{\text{top}}{C} = \frac{7}{16}$ and $L(z) = \frac{1}{1 + az^{-1}} = \boxed{\dfrac{1}{1 - \frac{3}{4}z^{-1}}}$.

Multiple choice "what must the innovation filter be?" → **minimum phase** (poles and zeros inside the unit circle — the label is the whole answer; Jun26 a).

### 1D · Expected value and the MC words

**IF YOU SEE "expected value", "discrete or continuous", "exactly $k$ of $n$", "covariance", "contour plot"**

- **Expected value = probability-weighted average.** Discrete: $E[x] = \sum_x x\,p(x)$. Continuous: $E[x] = \int x\,p(x)dx$ with the pdf's ordinary domain limits. For $E[x^2]$, put $x^2p(x)$ inside instead. With two variables, use the explicit slice limits from 1A and integrate $x\,p(x,y)$ (or $y\,p(x,y)$ for $E[y]$). The pdf alone sums/integrates to 1, **not** to the mean. **Mode** = most likely value. **Variance** $\mathrm{Var}(x) = E[x^2] - (E[x])^2$; standard deviation $= \sqrt{\mathrm{Var}}$.
- **Discrete** = the values can be listed: "number of ...", counts, dice, digits 0–255. **Continuous** = a measurement: time, distance, weight, money.
- **Rules:** $E[c\cdot z] = cE[z]$ · $E[c] = c \cdot E[z_1 + z_2] = E[z_1] + E[z_2]$ — a constant factor comes out. Uniform $[a,b] \to$ mean $\frac{a+b}{2}$ ($[-A, A] \to 0$).
- **"Exactly $k$ of $n$ tries"** (binomial): $C(n,k)\,p^k(1-p)^{n-k}$, $p$ = chance of one try, $C(n,k) = \frac{n(n-1)\cdots(n-k+1)}{k!}$. *e.g.* $n = 4$, $k = 2$, $p = 0.3$: $C(4,2) = \frac{4\cdot3}{2\cdot1} = 6 \to 6(0.3)^2(0.7)^2 = 0.2646$.
- **Covariance:** $\mathrm{Cov}(x,y) = E[xy] - E[x]E[y]$ · sign = whether one goes up when the other does · the number changes with units · correlation coefficient $= \mathrm{Cov}/(\sigma_x\sigma_y)$, always in $[-1,1]$. Independent → covariance 0; covariance 0 does **not always** prove independence. *Example:* $p(0,0) = p(1,1) = \frac{1}{2}$: $E[x] = E[y] = E[xy] = \frac{1}{2} \to \mathrm{Cov} = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$.
- **Contour plot, bi-variate Gaussian:** covariance = the two corner (off-diagonal) entries of the covariance matrix: 0 → ellipses aligned with the axes · **tilted** ellipses → covariance $\ne 0$.
- **Definition MC / T/F:** $F(t) = P(x \le t)$ (the cumulative), so $F(-\infty) = 0$, $F(\infty) = 1$ · independent $= P(A \mid B) = P(A)$, the same as $P(A \cap B) = P(A)P(B)$ ($\cap$ = "both happen") · **disjoint** = they can never both happen, $P(A \cap B) = 0$ (the opposite!) · Bayes = conditional probability rearranged. A T/F list → the swap test of box 2D, any topic.

## PART 2 · Estimation (3–3.5 p)

### 2A · Maximum likelihood (MLE)

**IF YOU SEE a pdf $p(x \mid \theta)$, samples $x_1..x_N$, "maximum likelihood", "derive the MLE", "find an estimator", "estimate of ..."**

For $N$ **iid** samples (independent, same pdf), $L(\theta) = \prod_{n=1}^{N} p(x_n \mid \theta)$: multiply one pdf by itself $N$ times, using $x_1,\dots,x_N$. Maximise $\ln L$ instead; $\ln$ turns multiplication into addition and does not move the maximum.

1. **Log:** $\boldsymbol{\ell(\theta) = \ln L = \sum_n \ln p(x_n \mid \theta)}$. Keep every term containing $\theta$. A constant term repeated for every sample becomes $N$ times that term.
2. **Differentiate with respect to $\theta$:** treat every $x_n$ as a known number; terms without $\theta$ vanish. Write $\ell'(\theta)$.
3. **Solve** $\ell'(\theta) = 0$, respecting the parameter's allowed range. Multiply through by denominators first if useful. The solution $\hat{\theta}$ is the MLE; if no stationary point exists, check allowed endpoints. A quick maximum check is $\ell''(\hat{\theta}) < 0$ when applicable.

**How to take $\ln$ of a pdf:** write the pdf as $\frac{\text{above}}{\text{below}} \cdot e^u$ (it is pieces multiplied, some in a fraction). Then each piece becomes one term: a piece $a$ **above the line** → $+\ln a$ · a piece $b$ **below the line** → $-\ln b$ · $e^u \to$ just $u$ (the $e$ disappears). Two pieces below the line split: $-\ln(2\theta) = -\ln 2 - \ln\theta$. A power $\theta^k$ below the line → $-k\ln\theta$. Derivatives: $\ln\theta \to \frac{1}{\theta}\cdot\theta^k \to k\theta^{k-1}\cdot\frac{1}{\theta} \to -\frac{1}{\theta^2}\cdot e^{c\theta} \to ce^{c\theta}\cdot\cos\theta \to -\sin\theta\cdot\sin\theta \to \cos\theta$. Solving $= 0$: multiply the equation by the denominators ($\theta^2$, $\theta$, ...) first — that does not change the solution. **Trap:** if the pdf only exists for $x$ up to something containing $\theta$ (uniform on $[0,\theta]$), these 3 steps are useless — $\hat{\theta} = \max_n x_n$, and **the CRLB does not exist**. **Only if Fisher information / CRLB is asked:** CRLB is the lowest possible variance of an unbiased estimator; reaching it = efficient. **(1)** Take $\ln p$ for ONE sample. **(2)** Differentiate with respect to the unknown twice. **(3)** Put a minus in front and take the expectation: $i(\theta) = -E[\text{second derivative of } \ln p]$. If $x$ remains, replace it using the known $E[x]$. **(4)** For $N$ iid samples, $\mathcal{I} = N\,i(\theta)$ and $\boxed{CRLB = 1/(N\,i(\theta))}$. *Exponential example,* $p = \lambda e^{-\lambda x}$ : $\ln p = \ln\lambda - \lambda x \to$ first derivative $= 1/\lambda - x \to$ second derivative $= -1/\lambda^2 \to i(\lambda) = -E[-1/\lambda^2] = 1/\lambda^2 \to CRLB = \lambda^2/N$. **Shortcut values of $i$:** Gaussian mean, known variance $\sigma^2$: $1/\sigma^2$ · exponential rate $\lambda$: $1/\lambda^2$ · Bernoulli $\theta$: $1/[\theta(1-\theta)]$ · Rayleigh $p = \frac{x}{\theta^2}e^{-x^2/2\theta^2}$: $4/\theta^2$, hence CRLB $= \theta^2/(4N)$.

**Check your $\hat{\theta}$:** uniform $[0,\theta] \to \max_n x_n$ (the trap above) · exponential rate $\lambda \to 1/\bar{x}$ · Poisson $\lambda \to \bar{x}$ · Bernoulli $\to k/N$ (= success in $N$).

*Apr24, follow the three steps.* $p(x \mid \theta) = \frac{x}{2\theta}e^{-x^2/(2\theta)}$. One sample: $\ln p = \ln x - \ln 2 - \ln\theta - \frac{x^2}{2\theta}$. Therefore $\ell = \sum \ln x_n - N\ln 2 - N\ln\theta - \frac{\sum x^2}{2\theta}$. Differentiate: $\ell' = -\frac{N}{\theta} + \frac{\sum x^2}{2\theta^2}$. Set to 0 and multiply by $2\theta^2$: $-2N\theta + \sum x_n^2 = 0 \to \boxed{\hat{\theta} = \frac{1}{2N}\sum x_n^2}$.

*Jun26* Gamma $\frac{x^{k-1}e^{-x/\theta}}{\Gamma(k)\theta^k}$, $n$ patients: $\ln p = (k-1)\ln x - \frac{x}{\theta} - k\ln\theta - \ln\Gamma(k)$. Add all $n$ logs: $\ell = (k-1)\sum \ln x_i - \frac{\sum x_i}{\theta} - nk\ln\theta - n\ln\Gamma(k)$. Differentiate and drop terms without $\theta$: $\ell' = \frac{\sum x_i}{\theta^2} - \frac{nk}{\theta} = 0$. Multiply by $\theta^2$: $\sum x_i - nk\theta = 0 \to \boxed{\hat{\theta} = \frac{\sum x_i}{nk} = \frac{\bar{x}}{k}}$.

**Sigmoid trick.** IF YOU SEE $\lambda = \frac{1}{1+e^{-\theta}}$: treat $\lambda$ as the unknown, find $\hat{\lambda}$, then convert with $\hat{\theta} = -\ln\left(\frac{1}{\hat{\lambda}} - 1\right)$.

*Apr25, one observation:* $p = 3\lambda e^{-4\lambda x/7} \to \ln p = \ln 3 + \ln\lambda - \frac{4\lambda x}{7} \to$ derivative $\frac{1}{\lambda} - \frac{4x}{7} = 0 \to \frac{1}{\lambda} = \frac{4x}{7} \to \hat{\lambda} = \frac{7}{4x}$. Convert: $\hat{\theta} = -\ln(1/\hat{\lambda} - 1) = -\ln(\frac{4x}{7} - 1)$, valid when $x > \frac{7}{4}$.

### 2B · MAP with a prior (Beta / Gamma)

**IF YOU SEE "prior", "posterior", "MAP", "maximum a posteriori"/"a-posteriori", "MMSE"/"minimum mean square error", "Bayesian estimator", Beta, Gamma**

**Central rule:** posterior $\propto$ likelihood $\times$ prior. **MAP** = parameter value at the posterior's maximum; **MMSE** = posterior mean. Multiplicative constants not containing the unknown do not change MAP.

1. **Likelihood:** write each observation's probability, keep only factors containing the unknown, multiply them; equal bases → add exponents.
2. **Multiply by the prior**, adding exponents again. Call the result $q(\theta)$. It may be unnormalised (its integral need not be 1).
3. Match $q$ to a row below: $A, B$ are the FINAL exponents after including every observation and the prior.

| Result looks like | Posterior | MAP | MMSE (mean) |
|---|---|---|---|
| $\theta^A(1-\theta)^B$ | Beta$(A+1, B+1)$ | $\frac{A}{A+B}$ | $\frac{A+1}{A+B+2}$ |
| $\lambda^A e^{-B\lambda}$ | Gamma$(A+1, B)$ | $\frac{A}{B}$ | $\frac{A+1}{B}$ |

Prior Beta$(a,b) = \theta^{a-1}(1-\theta)^{b-1}$ (its mean $\frac{a}{a+b}$). Prior Gamma$(a,b) = \lambda^{a-1}e^{-b\lambda}$ (mean $\frac{a}{b}$). The prior pulls the estimate towards that mean. One observation $x$ gives: Poisson count → $\lambda^x e^{-\lambda}$ · "$k$-th success on trial $x$" → $\theta^k(1-\theta)^{x-k}$ · "$k$ successes in $n$ tries" → $\theta^k(1-\theta)^{n-k}$ · exponential → $\lambda e^{-\lambda x}$.

**If your result is not in the table:** MAP = the allowed $\theta$ where $q$ is largest; usually solve $\frac{d}{d\theta}\ln q = 0$. MMSE = $\dfrac{\int \theta q(\theta)d\theta}{\int q(\theta)d\theta}$ using the allowed parameter limits. The denominator is required: it makes the posterior's total probability 1. If the best point is outside the prior's allowed interval, choose the nearest endpoint for MAP.

*Jun26-2c (Beta):* each observation contributes $\theta^2(1-\theta)^{x-2}$.

**Step 1 — multiply the five likelihood factors:** $\theta$ powers add: $2+2+2+2+2 = 10$. The $(1-\theta)$ powers add: $(3-2)+(5-2)+(4-2)+(6-2)+(4-2) = 12$. Likelihood $= \theta^{10}(1-\theta)^{12}$.

**Step 2 — multiply the prior:** Beta(2,3) contributes $\theta^{2-1}(1-\theta)^{3-1} = \theta^1(1-\theta)^2$. Final powers: $q = \theta^{11}(1-\theta)^{14}$.

**Step 3 — read the table with** $A = 11$, $B = 14$**:** MAP $= \frac{11}{11+14} = \frac{11}{25} = 0.44$; MMSE $= \frac{12}{11+14+2} = \frac{12}{27} = 0.444$.

*Jun26-2d (Gamma), Poisson observations* $2, 4, 3, 5, 2, 4$*:*

**Step 1 — likelihood:** $\lambda$ powers add: $2+4+3+5+2+4 = 20$. Six observations give six factors $e^{-\lambda}$, so likelihood $= \lambda^{20}e^{-6\lambda}$.

**Step 2 — multiply prior Gamma(3,2):** prior $= \lambda^{3-1}e^{-2\lambda} = \lambda^2 e^{-2\lambda}$. Add powers and exponential coefficients: $q = \lambda^{22}e^{-8\lambda}$.

**Step 3 — read the table with** $A = 22$, $B = 8$**:** MAP $= \frac{22}{8} = 2.75$; MMSE $= \frac{23}{8} = 2.875$. Without the prior, MLE $= \frac{20}{6} = 3.33$; the prior pulls toward its mean $\frac{3}{2}$.

*Nov22:* unnormalised $q = \lambda^{-(x+a)\lambda} \to$ MAP $= 1/(x+a)$; MMSE uses the normalising ratio $= 2/(x+a)$ (fully integrated in Calculations). For a sigmoid parameter, use the conversion at the end of 2A.

### 2C · Least-squares line

**IF YOU SEE points $(x,y)$ + "least squares"/"least-squares"/"LSE", "linear function", "fit a line"**

1. Means $\bar{x}$ and $\bar{y}$ (bar = average).
2. Slope: $a = \dfrac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2}$.
3. Intercept: $b = \bar{y} - a\bar{x}$. Answer: $y = ax + b$. Check: $x = \bar{x}$ must give $y = \bar{y}$.

*Apr25, points* $(0,3), (2,4), (3,3)$*:*

**Step 1 — averages:** $\bar{x} = \frac{0+2+3}{3} = \frac{5}{3}$ and $\bar{y} = \frac{3+4+3}{3} = \frac{10}{3}$.

**Step 2 — subtract the average from EACH point:** $x - \bar{x} = (-\frac{5}{3}, \frac{1}{3}, \frac{4}{3})$ and $y - \bar{y} = (-\frac{1}{3}, \frac{2}{3}, -\frac{1}{3})$.

**Step 3 — form the two sums:** numerator $= (-\frac{5}{3})(-\frac{1}{3}) + (\frac{1}{3})(\frac{2}{3}) + (\frac{4}{3})(-\frac{1}{3}) = \frac{5+2-4}{9} = \frac{3}{9}$. Denominator $= (-\frac{5}{3})^2 + (\frac{1}{3})^2 + (\frac{4}{3})^2 = \frac{25+1+16}{9} = \frac{14}{3}$.

**Step 4 — slope, then intercept:** $a = \frac{1/3}{14/3} = \frac{1}{14}$; $b = \bar{y} - a\bar{x} = \frac{10}{3} - \frac{1}{14}\frac{5}{3} = \frac{45}{14}$. Therefore $\boxed{y = \frac{1}{14}x + \frac{45}{14}}$. Check: putting $x = \bar{x}$ gives $y = \bar{y}$.

Also: $x[n] = \theta + \text{noise} \to \hat{\theta} = \bar{x}$. $x[n] = \ln\theta + \text{noise} \to \hat{\theta} = e^{\bar{x}}$ (noise = random wobble; fitting a constant by least squares is just its average).

**Beyond a straight line.** $J(\theta) = \sum_n (x[n] - s[n;\theta])^2$, $\hat{\theta} = \arg\min J$ (the $\theta$ making $J$ smallest) — write this when they ask "what kind of estimator". One amplitude $s[n] = Ah[n]$: $\hat{A} = \dfrac{\sum x[n]h[n]}{\sum h[n]^2}$ (*e.g.* $x = 3, 5$, $h = 1, 2$: $\hat{A} = \frac{3+10}{1+4} = 2.6$).

Linear model $x = H\theta + w$: $\hat{\theta} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T x$; the **error** $x - H\hat{\theta}$ is orthogonal (perpendicular) to the columns of $\mathbf{H}$ (it is the error, **not** $\hat{\theta}$); weighted LS with $\mathbf{W} = \mathbf{C}_x^{-1} =$ **BLUE** (best linear unbiased estimator). Nonlinear: transform it (the $e^x$ line), or iterate **Gauss–Newton** $\theta_{k+1} = \theta_k + (\mathbf{G}^T\mathbf{G})^{-1}\mathbf{G}^T(x - s(\theta_k))$, $\mathbf{G}[:,i] = \frac{\partial s}{\partial \theta_i}$ — Newton–Raphson works too, "only Gauss–Newton" is **false**.

### 2D · Part 2 true / false

**IF YOU SEE "which of the following statements" (in Part 2) about MLE, MAP, MMSE, Fisher, CRLB**

**T/F method:** translate each option into an equation before deciding. A common false option is a true sentence with ONE word swapped: hunt *always · regardless · strictly · maximise/minimise · mean/maximum*. This is a warning sign, **not proof**; "always" is killed by one counterexample, and a true sentence needs no swap word.

| The fact (true) | The lie they print (false) |
|---|---|
| MMSE = **mean** of the posterior; MAP = **maximum** | the two swapped |
| MMSE = MAP for a symmetric one-peak posterior | "**always** different" / "always equal" |
| MAP = likelihood × prior, needs a prior (Jun26 c) | "MAP→MLE **regardless of** the prior" · "Bayesians skip priors" |
| Bayes estimate **minimises** the Bayes risk (Nov22 b) | "maximises" |
| Quadratic cost → MMSE; uniform cost → MAP | "uniform cost → MMSE" |
| MLE = argmax of the likelihood **over** $\theta$ | "over $x$" · "only linear models" |
| MLE is **asymptotically** unbiased and efficient | "**always** reaches the CRLB" |
| Invariance: $\alpha = \psi(\theta)$ 1-to-1 → $\hat{\theta} = \psi^{-1}(\hat{\alpha})$ (Jan22 c) | — |
| Fisher of $N$ iid (independent, same pdf) $= N\,i(\theta)$, $\mathcal{I} \ge 0$; $\mathrm{Var}(\hat{\theta}) \ge \frac{1}{\mathcal{I}(\theta)}$ | "$\mathcal{I}$ strictly negative" · "$\ge \mathcal{I}(\theta)$" |
| CRLB $= \frac{1}{N\mathcal{I}(\theta)}$ shrinks $\sim \frac{1}{N}$ (Jan22 b) | "**always** derivable" (no, if range of $x$ depends on $\theta$) |
| efficient = variance **equal to** the CRLB | "**below** the CRLB" |
| Fisher info for "$x$ = signal($\theta$) + Gaussian noise $\sigma^2$": $\mathcal{I} = \frac{1}{\sigma^2}\sum\left(\frac{d\,\text{signal}}{d\theta}\right)^2$. | |

*Nov22* $A[\cos\theta, \sin\theta]$: **phase MLE** $= \arctan\frac{x_2}{x_1}$ (point the model vector at the data), $\mathcal{I} = \frac{A^2}{\sigma^2}$.

## PART 3 · Spectral estimation (3 p)

$r[k]$ = autocorrelation at lag $k$ (the given numbers; lag $k$ = $k$ steps apart). Rewrite cosines: $\cos 2\theta = 2\cos^2\theta - 1$, $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, $\cos\alpha\cos\beta = \frac{\cos(\alpha+\beta)+\cos(\alpha-\beta)}{2}$ (to expand $|L|^2$ of box 1C).

**Draw it** — the question says "determine **and plot**", only the numbers are graded: $r[k] \to$ vertical **stems** (one stick per lag) at the lags. $P(\theta) \to$ compute the value at $\theta = 0$ and at $\theta = \pi$, draw **one smooth curve** through those two heights and write them on the axis. Peak at 0 = low-pass (low frequencies), peak at $\pi$ = high-pass, peak in between = band-pass.

### 3A · Correlogram

**IF YOU SEE $r[0], r[1], r[2]\ldots$ + "correlogram", "only these lags", "PSD as a function of $\cos\theta$", correlation values given**

$P(\theta) = r[0] + 2r[1]\cos\theta + 2r[2]\cos(2\theta) + 2r[3]\cos(3\theta) + \cdots$. Stop at the largest given lag. **Why the 2?** Correlation at lag $+k$ equals correlation at lag $-k$; their two equal contributions combine into $2r[k]\cos(k\theta)$. The zero lag occurs only once, so **do not double** $r[0]$. From a data list, $\hat{r}[k] = \frac{1}{N}$(sum of products of values $k$ positions apart): use only pairs that both exist, but still divide by the full data length $N$. Reverse question ($P$ given): $r[0]$ = constant term; $r[k]$ = HALF the coefficient of $\cos(k\theta)$. Check your expression at $\theta = 0$ (every cosine = 1) and $\theta = \pi$ (cosines alternate between $-1$ and 1).

*Jun26* 3, 2, 1: $3 + 4\cos\theta + 2\cos 2\theta = 4\cos^2\theta + 4\cos\theta + 1 = (2\cos\theta + 1)^2$. *From data* $x = [1, 2, 3]$: $\hat{r}[0] = \frac{1+4+9}{3} = \frac{14}{3}$, $\hat{r}[1] = \frac{2+6}{3} = \frac{8}{3}$ (only the 2 overlapping pairs), $\hat{r}[2] = \frac{3}{3} = 1$.

## 3B · AR(1)

**IF YOU SEE "AR(1)", "first-order autoregressive"**

Use the LEFT-side convention $x[n] + a\,x[n-1] = i[n]$ (equivalently $x[n] = -a\,x[n-1] + i[n]$), with white-noise variance $\sigma^2$. The sign of $a$ must match this equation.

1. $a = -\dfrac{r[1]}{r[0]}$
2. $\sigma^2 = r[0] + a\,r[1]$
3. $P = \dfrac{\sigma^2}{1 + a^2 + 2a\cos\theta}$ (check $|a| < 1$ and $\sigma^2 > 0$, else a sign went wrong)

*Apr25* $r[0] = 2$, $r[1] = -\frac{1}{4}$: $a = -\frac{-1/4}{2} = \frac{1}{8}$ · $\sigma^2 = 2 + \frac{1}{8}(-\frac{1}{4}) = \frac{63}{32}$ → $P = \frac{63/32}{1 + \frac{1}{64} + \frac{1}{4}\cos\theta}$ · $\frac{64}{64} = \frac{126}{65 + 16\cos\theta}$.

## 3C · AR(2) / Yule–Walker

**IF YOU SEE "AR(2)", "Yule–Walker"**, $r[0], r[1], r[2]$

Convention: $x[n] + a_1 x[n-1] + a_2 x[n-2] = i[n]$. Substitute the three given correlations in order:

1. $D = r[0]^2 - r[1]^2$ (shared denominator)
2. $a_1 = \dfrac{-r[1]\,(r[0] - r[2])}{D}$, $a_2 = \dfrac{r[1]^2 - r[0]\,r[2]}{D}$
3. $\sigma^2 = r[0] + a_1 r[1] + a_2 r[2]$
4. $P = \dfrac{\sigma^2}{1 + a_1^2 + a_2^2 + 2a_1(1 + a_2)\cos\theta + 2a_2\cos 2\theta}$

If they want the matrix (Yule–Walker = steps 1–2 in matrix form): $\begin{bmatrix} r[0] & r[1] \\ r[1] & r[0] \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = -\begin{bmatrix} r[1] \\ r[2] \end{bmatrix}$.

*Jun26* $r = 3, 2, 1$: $D = 3^2 - 2^2 = 5$ · $a_1 = \frac{-2(3-1)}{5} = -\frac{4}{5}$ · $a_2 = \frac{2^2 - 3\cdot 1}{5} = \frac{1}{5}$ · $\sigma^2 = 3 + (-\frac{4}{5})2 + \frac{1}{5}(1) = \frac{8}{5}$ → step 4, then $\cos 2\theta = 2\cos^2\theta - 1$ → $P = \frac{10}{5\cos^2\theta - 12\cos\theta + 8}$.

## 3D · MA(1)

**IF YOU SEE "MA(1)", "moving average"**

Model: $x[n] = i[n] + b\,i[n-1]$. PSD: $P = r[0] + 2r[1]\cos\theta$ — it uses $r[0], r[1]$ only: ignore $r[2]$ and beyond. If they ask for $b$ and $\sigma^2$: solve $b^2 - \frac{r[0]}{r[1]}\,b + 1 = 0$, keep the root between $-1$ and 1; then $\sigma^2 = \frac{r[1]}{b}$.

*Apr25* $r = 2, -\frac{1}{4}$: $P = 2 - \frac{1}{2}\cos\theta$. $b, \sigma^2$ asked too: $b^2 - \frac{r[0]}{r[1]}b + 1 = b^2 + 8b + 1 = 0 \to b = -4 + \sqrt{15} = -0.127$ (the root with $|b| < 1$), $\sigma^2 = \frac{r[1]}{b} = \frac{4 + \sqrt{15}}{4} = 1.97$; check $r[0] = \sigma^2(1 + b^2) = 2$ ✓

## 3E · Model from an equation

**IF YOU SEE** $x[n] = \ldots$ **+ "which kind of process", "calculate the autocorrelation of** $x$**"**

**1 · Name the model by looking only at the right-hand side.** $i$ means input noise. Only noise terms → **MA**; the largest noise delay is $q$. Noise AND earlier $x$ terms → **ARMA**; the largest earlier-$x$ delay is $p$. Earlier $x$ terms with only $i[n]$ → **AR**: use 3B/3C. If asked from a PSD shape: sharp peaks → AR, deep nulls → MA, both → ARMA.

**2 · MA autocorrelation: copy coefficients, then multiply pairs.** For $x[n] = i[n] + b_1 i[n-1] + b_2 i[n-2] + \cdots$, write the coefficient list $(1, b_1, b_2, \ldots)$ and noise variance $\sigma^2$. $r[0] = \sigma^2$(sum of EVERY coefficient squared). $r[1] = \sigma^2$(sum of products of neighbouring coefficients). $r[2] = \sigma^2$(sum of products two positions apart), and so on. After the largest delay, $r[k] = 0$.

*Apr24, every pair shown:* coefficients $(1, -\frac{1}{2}, \frac{1}{4}, -\frac{1}{2})$, $\sigma^2 = 1$.
$r[0] = 1^2 + (-\frac{1}{2})^2 + (\frac{1}{4})^2 + (-\frac{1}{2})^2 = \frac{25}{16}$.
$r[1] = (1)(-\frac{1}{2}) + (-\frac{1}{2})(\frac{1}{4}) + (\frac{1}{4})(-\frac{1}{2}) = -\frac{3}{4}$.
$r[2] = (1)(\frac{1}{4}) + (-\frac{1}{2})(-\frac{1}{2}) = \frac{1}{2}$ · $r[3] = (1)(-\frac{1}{2}) = -\frac{1}{2}$.

**Backwards (coefficients asked from** $r[k]$**):** for MA(2), start at the largest lag: $b_2 = r[2]/\sigma^2$; then $r[1] = \sigma^2 b_1(1 + b_2)$, so $b_1 = r[1]/[\sigma^2(1 + b_2)]$. Finally $r[0] = \sigma^2(1 + b_1^2 + b_2^2)$.

**3 · ARMA(1,1): move earlier** $x$ **to the LEFT.** Match $x[n] + \alpha x[n-1] = i[n] + \beta i[n-1]$; the signs come from this left-side form. Then $P = \sigma^2\dfrac{1 + \beta^2 + 2\beta\cos\theta}{1 + \alpha^2 + 2\alpha\cos\theta}$. If correlations are asked: $r[0] = \sigma^2\dfrac{1 + \beta^2 - 2\alpha\beta}{1 - \alpha^2}$, $r[1] = -\alpha r[0] + \beta\sigma^2$, and each following $r[k]$ is the previous one times $-\alpha$.

*Nov22:* $x[n] = i[n] - i[n-1] + \frac{3}{4}x[n-1]$ becomes $x[n] - \frac{3}{4}x[n-1] = i[n] - i[n-1]$, so $\alpha = -\frac{3}{4}$, $\beta = -1$. With $\sigma^2 = \frac{7}{8}$: numerator $= \frac{7}{8}(1 + 1 - 2\cos\theta)$, denominator $= 1 + \frac{9}{16} - \frac{3}{2}\cos\theta \to P = \dfrac{28(1 - \cos\theta)}{25 - 24\cos\theta}$.

## 3F · Pick the plot

**IF YOU SEE "which plot", "which figure", "plots A–D", "match the plot"**

Work out each PSD at $\theta = 0$ (all cosines = 1) and at $\theta = \pi$ ($\cos\theta = -1$, $\cos 2\theta = 1$, $\cos 3\theta = -1$). Pick the plot with those heights. Only a correlogram can go below 0.

*Apr25:* correlogram 1.13 → 2.13, MA 1.5 → 2.5, AR 1.56 → 2.57 (heights at $0 \to \pi$) → plot D. **Second type** (windowed cosine, DTFT vs DFT — the DTFT is the whole curve, the DFT is the dots on it): infinite cosine = **2 spikes** · windowed = **main lobe + side lobes** (one big bump + small ones) · DFT = **samples of that same curve** · zero padding = **more dots**, not more resolution.

## 3G · Welch / Bartlett count

**IF YOU SEE "Welch"/"Bartlett"/"WOSA",** $N$ **samples, segment length** $L$**, overlap %, "how many segments"**

**Convert overlap to a fraction first** (50 % → 0.5). Step/start spacing $D = L(1 - \text{overlap})$. Segment starts are $0, D, 2D, \ldots$; include a start only if its whole length-$L$ segment fits. Hence $K = \left\lfloor \frac{N - L}{D} \right\rfloor + 1$. Bartlett = 0 overlap, so $K = \lfloor N/L \rfloor$. Ideal independent-segment variance ratio $= 1/K$; real overlapping Welch segments are correlated, so reduction is less.

*Jun26* $N = 1000$, $L = 200$, 50 %: $D = 200(1 - 0.5) = 100$, $K = \frac{1000 - 200}{100} + 1 = 9$ segments, ratio $\frac{1}{9}$.

## 3H · Energy or power signal

**IF YOU SEE "energy or power signal", "total energy", "average power"**

- Dies out or has finite length → **energy signal**: $0 < E = \sum_n |x[n]|^2 < \infty$, average power $= 0$.
- Repeats/never dies (sin, cos, nonzero constant, $\cos t\,u(t)$) → **power signal**: $E = \infty$, finite nonzero $P$ = time average of $|x|^2$. A sinusoid $A\sin(\omega t)$ or $A\cos(\omega t)$ has $P = A^2/2$.
- Continuous time: $E = \int_{-\infty}^{\infty} |x(t)|^2 dt$; for a periodic signal, $P = \frac{1}{T}\int_{t_0}^{t_0+T} |x(t)|^2 dt$ over ONE period $T$. Discrete periodic: $P = \frac{1}{N}\sum_{n=0}^{N-1} |x[n]|^2$.

*Apr25:* $A\sin\omega t$ → power $\frac{A^2}{2}$. $e^{-n}$ for $n \geq 0$ → energy $\sum e^{-2n} = \frac{1}{1 - e^{-2}} \approx 1.16$ (use $\sum_{n\geq 0} q^n = \frac{1}{1-q}$). *Continuous* $e^{-t}$, $t \geq 0$: dies out → energy $\int_0^\infty e^{-2t} dt = \frac{1}{2}$ (the $c = 2$ line of the Calculations strip).

## 3I · Name the method

**IF YOU SEE a table of descriptions to fill with method names, "name the method", bias/variance to match**

| The description says… | Method |
|---|---|
| $\lvert\text{DFT}\rvert^2$ divided by $N$, whole signal | Periodogram |
| DFT of the estimated autocorrelation | Correlogram |
| DFT of the estimated autocorrelation, then truncate/window it at maximum lag $M < N$; "consistent" | Blackman–Tukey |
| NON-overlapping segments, average the periodograms | Bartlett |
| OVERLAPPING segments, average the periodograms | Welch (WOSA) |
| $\sigma^2\lvert B\rvert^2/\lvert A\rvert^2$, "linear random signal model" | Parametric (AR / MA / ARMA) |
| "averages periodograms" | Bartlett and Welch |

**How:** hunt ONE trigger word per description — even reworded, the trigger stays.

## 3J · Part 3 true / false

**IF YOU SEE "which of the following statements" (in Part 3) about PSD, windows, periodogram, Welch**

Same swap test as 2D — hunt: *improves/reduces · exactly · more/less · both/only · resolution↔leakage*.

| The fact (true) | The lie they print (false) |
|---|---|
| Periodogram is **not** consistent: bias → 0 but $\text{Var}\{\hat{P}\} \approx P^2$ stays (Jun26 b) | "it is consistent" |
| Bias affects **both** deterministic and stochastic signals (Apr25 a) | "only stochastic" |
| Bartlett: variance $\propto \frac{1}{K}$, bias **up** (Nov22 a) | "variance drops by zero-padding" (it splits the data) |
| Welch (overlapping): variance drops **less** than $\frac{1}{K}$ | "exactly $\frac{1}{K}$" |
| Main-lobe width → resolution · side-lobe height → leakage (Nov22 b) | the two swapped |
| Windows (Hamming…) cut **leakage**; rectangular has best **resolution** | "a window improves resolution" · "triangular beats rectangular" |
| Zero-padding: more points (step $\frac{2\pi}{P}$), **no** better resolution | "improves resolution" |
| Window type does **not** change the periodogram's variance; leakage follows the window, not the record length | "longer data/window reduces leakage" |
| Periodogram and correlogram have the **same** bias | "periodogram has better bias" |

*Consistent* = variance → 0 as $N \to \infty$. The periodogram's bias → 0 but $\text{Var}\{\hat{P}\} \approx P^2$ stays put.

**AR order (Jun26-3e).** Bigger $p$ → residual error always goes down → risk: **overfitting** (fake peaks). Fix: **AIC** = $N\ln\hat{\sigma}^2 + 2p$ (also FPE, MDL): the error term falls, the penalty $2p$ grows → choose the $p$ with the smallest AIC. Any PSD can be approximated by an AR model of **high enough order**.

# PART 4 · Detection (2–2.5 p)

$H_0$ = nothing there, $H_1$ = something there. $P_{FA}$ = decide $H_1$ when $H_0$ is true (false alarm). $P_D$ = decide $H_1$ when $H_1$ is true. $P_M = 1 - P_D$ (miss).

## 4A · Likelihood-ratio test

**IF YOU SEE two pdfs (one for** $H_0$**, one for** $H_1$**) + "binary hypothesis"/"hypothesis testing", "likelihood ratio test", "when accept** $H_1$**", threshold** $\lambda, \gamma, \eta$**, "**$P_{FA}$**"/"**$P_D$**"**

1. Put $H_1$ ON TOP: $L(\mathbf{x}) = p(\mathbf{x} \mid H_1)/p(\mathbf{x} \mid H_0)$. For independent samples, multiply the per-sample ratios; equal exponentials combine by adding exponents. Cancel common factors.
2. Start with $L(\mathbf{x}) > \eta \Rightarrow$ decide $H_1$. Take ln on both sides (allowed because ln is increasing): $\ln L > \ln\eta$. Expand logs BEFORE isolating the statistic.
3. Isolate $x$, $\sum x_n$, $\sum |x_n|$, or $\sum x_n^2$. Multiplying/dividing by a negative flips $>$ to $<$; square roots need the stated domain.
4. **Check the decision side numerically:** choose one simple value on each side, evaluate $L$, and mark where $L > \eta$. Then compute error areas on those sides. Equality can be assigned either way for continuous pdfs (probability 0).

*Apr25* $\frac{6e^{-1}}{3e^{-2}} = 2e^6 > 0.2 \to x^6 > 0.1 \to x > 0.1^{1/6} = 0.68$.

*Jun26 Rayleigh:* $p(x \mid H_j) = \frac{x}{\sigma_j^2} e^{-x^2/(2\sigma_j^2)}$, $x \geq 0$, $\sigma_1 > \sigma_0$.

**Step 1 — divide** $p_1$ **by** $p_0$: $x$ cancels, leaving $L = \frac{\sigma_0^2}{\sigma_1^2}\exp[\frac{x^2}{2}(\frac{1}{\sigma_0^2} - \frac{1}{\sigma_1^2})]$.

**Step 2 — use** $L > \eta$ **and take ln:** $\frac{x^2}{2}(\frac{1}{\sigma_0^2} - \frac{1}{\sigma_1^2}) > \ln(\eta\frac{\sigma_0^2}{\sigma_1^2})$.

**Step 3 — isolate** $x$**:** the bracket is positive, so no inequality flip. Multiply by its reciprocal and take the positive square root: $\boxed{x > \sqrt{\dfrac{2\sigma_0^2\sigma_1^2}{\sigma_1^2 - \sigma_0^2}\ln(\eta\dfrac{\sigma_1^2}{\sigma_0^2})}}$. Larger separation between the pdfs → better detection.

*Apr24* Gaussian $\frac{1}{\sqrt{2\pi}}e^{-y^2/2}$ vs exponential $\frac{1}{4}e^{-y/2}$, $\gamma = 1$: the constant $\ln\frac{1/4}{1/\sqrt{2\pi}} = \ln\frac{4}{2.51} = -0.467 \to -0.467 + \frac{y^2}{2} - \frac{y}{2} > 0$ → $y^2 - y - 0.934 > 0$ → root $y = \frac{1 + \sqrt{1 + 3.74}}{2} = 1.59$ → accept $H_1$ if $y > 1.59$. $P_{FA} = Q(1.59) = 0.056$, $P_D = \frac{1}{2}e^{-0.795} = 0.23$ (printed answer — that pdf is unnormalised, keep the $\frac{1}{2}$).

*Nov22* $\frac{1}{2}e^{-|x|}$ vs $e^{-2|x|}$, $N$ samples: $L = 2^N e^{-\sum|x_n|}$ → accept $H_1$ if $\sum|x_n| < N\ln 2 - \ln\lambda$.

**Probabilities.** $P_{FA}$ = area under the $H_0$ pdf where you decide $H_1$. $P_D$ = area under the $H_1$ pdf in the same place. Tail formulas: Gaussian $X \sim N(\mu, \sigma^2)$: $P(X > t) = Q((t - \mu)/\sigma)$; standard Gaussian $\mu = 0, \sigma = 1 \to Q(t)$ · exponential $ce^{-cx}$, $x \geq 0$: $P(X > t) = e^{-ct}$ and $P(X < t) = 1 - e^{-ct}$ · Rayleigh: $P(X > t) = e^{-t^2/(2\sigma^2)}$ · symmetric two tails $|X| > t \to$ twice one tail. **Always use the distribution under the hypothesis named by the error:** $P_{FA}$ uses $H_0$; $P_D$ uses $H_1$.

"Threshold for $P_{FA} = 0.1$": set the $H_0$ area $= 0.1$, solve for $t$. **Read** $Q$ **backwards:** with a Gaussian $H_0$ the $t$ with $Q(t) = 0.1$ is $t = 1.28$ (Numbers strip); then $P_D$ = area under the $H_1$ pdf from that same $t$ on.

*Nov22:* $1 - e^{-t} = 0.1 \to t = -\ln 0.9$; then $P_D = 1 - e^{-2t} = 0.19$.

"Plot / indicate $P_{FA}$ and $P_M$": draw both pdfs and a vertical line at the threshold. Shade the $H_0$ curve on the "decide $H_1$" side → $P_{FA}$. Shade the $H_1$ curve on the "decide $H_0$" side → $P_M$. (Nov22: $H_1$ when $y < 1$, so $P_{FA} = H_0$ area left of 1, $P_M = H_1$ area right of 1.)

**Gaussian with any** $\sigma$. $p(x) = \dfrac{1}{\sigma\sqrt{2\pi}}e^{-x^2/2\sigma^2}$ (mean 0). Two zero-mean Gaussians, $\sigma_1$ vs $\sigma_0$ (pe21-4b): $\ln L = N\ln\frac{\sigma_0}{\sigma_1} + \frac{1}{2}(\frac{1}{\sigma_0^2} - \frac{1}{\sigma_1^2})\sum_n x^2[n] \to$ decide $H_1$ when $\sum_n x^2[n]$ is **big** = **energy detector** (bigger variance = more energy).

**erf table** (printed with the question — pe21-4a). The printed formula $\frac{1}{2}[1 + \text{erf}\,\frac{t-\mu}{\sigma\sqrt{2}}]$ gives $P(x < t)$, so $P(x > t) = 1 -$ that. First compute $z = \frac{t-\mu}{\sigma}$, then read erf $\frac{z}{\sqrt{2}}$ at the column $x = z$ ($x$ runs −3 to 3; the value at $-z$ is minus the value at $z$).

**Matched filter** (jan22-4a): correlate the data with the known signal $s[n]$ (FIR filter $h[n] =$ the **time-reversed** $s$) → biggest output SNR; "only for uncorrelated noise" is **false**.

## 4B · Priors and costs (Bayes)

**IF YOU SEE priors** $P(H_0), P(H_1)$**, costs** $C_{ij}$**, "posterior", "Bayesian hypothesis test", "**$P_M$**" / "miss"**

$C_{ij}$ = cost of deciding $i$ when $j$ is true. $C_{10}$ = false alarm cost, $C_{01}$ = miss cost. "Minimises the Bayes risk" = makes the total cost $C_{00}P(H_0)(1 - P_{FA}) + C_{10}P(H_0)P_{FA} + C_{11}P(H_1)P_D + C_{01}P(H_1)P_M$ as small as possible.

- Threshold $\eta = \dfrac{P(H_0)\,(C_{10} - C_{00})}{P(H_1)\,(C_{01} - C_{11})}$. Equal error costs and 0 for correct → $\eta = \frac{P(H_0)}{P(H_1)}$. Then do 4A with this $\eta$.
  *Jun26-4c:* $\eta = \frac{0.9}{0.1} = 9$ → put $\eta = 9$ in the Rayleigh answer of 4A.
- **Posteriors given** → accept $H_1$ if $\dfrac{P(H_1 \mid \mathbf{x})}{P(H_0 \mid \mathbf{x})} > \dfrac{C_{10} - C_{00}}{C_{01} - C_{11}}$.
  *Apr25-4c:* $C_{01} = 100\,C_{10}$ → right side $= 0.01$. Posterior ratio $= 5 > 0.01$ → accept $H_1$ → the student fails: **yes**.
- **"Where** $p_1 P_1 = p_0 P_0$**"** ($p$ = the two pdfs, $P$ = the two priors) → write the equality and solve.
  *Nov22:* $0.6 \cdot 2e^{-2y} = 0.4\,e^{-y} \to e^{-y} = \frac{1}{3} \to y = \ln 3$; decide $H_1$ for $y < \ln 3$. $P_{FA} = 1 - e^{-\ln 3} = \frac{2}{3}$, $P_M = e^{-2\ln 3} = \frac{1}{9}$.

## 4C · Part 4 true / false

**IF YOU SEE "which of the following statements" (in Part 4) about hypothesis testing,** $P_{FA}$**, errors**

Same swap test as 2D — the verb is the question: risk **min** · $P_D$ **max** · $P_e$ **min** ($P_e$ = total error $= P(H_0)P_{FA} + P(H_1)P_M$, equal costs) · $\eta$ goes **up** with $P(H_0)$, **down** with $P(H_1)$.

| The fact (true) | The lie they print (false) |
|---|---|
| Bayes detection **minimises** the Bayes risk | "maximises" |
| $C_{01} = C_{10}$, $C_{00} = C_{11} = 0$ → minimises total error $P_e$ | *trap:* scaling all costs leaves $\eta$ unchanged |
| Equal priors + equal costs → maximum-likelihood rule | — |
| Higher $P(H_0)$ raises $\eta$ → harder to pick $H_1$ | "higher $P(H_1)$ raises $\eta$" (lowers — Jun26 c) |
| Neyman–Pearson: fix $P_{FA}$, **maximise** $P_D$ | "minimise $P_D$" · "fix $P_D$, max $P_{FA}$" |
| NP's solution is the LRT; the LRT itself needs **no** priors | "the LRT needs $P_0, P_1$" |
| ln of the ratio never changes the decisions | "changes the decisions" |
| Errors are unavoidable: both prob overlap | "a threshold kills both errors" |
| ROC = $P_D$ plotted against $P_{FA}$ as the threshold moves: up and right · diagonal = guessing · better = top-left | "below diagonal = wrong $\sigma$" (flipped) |
| miss = false negative = type II · false alarm = false positive = type I (Nov22) | labels swapped |

Matched filter & energy-detector sentences → the last blocks of 4A: time-reversed $s$ · "only for uncorrelated noise" **false** · $\sum x^2$ **big** · a narrow $H_1$ pdf → the sum comes out **small**.

## Calculations · the algebra under every answer

**Integrals.** These formulas are **printed in the question itself** (q3-23-2h, nov22-2f) — you copy them: $\int e^{cx}dx = \frac{1}{c}e^{cx}$ · $\int xe^{cx}dx = \frac{cx-1}{c^2}e^{cx}$ · $\int x^2e^{cx}dx = (\frac{c^2}{c^3}x^2 - \frac{2x}{c^3} + \frac{2}{c^3})e^{cx}$ · $\int x^2e^{-x}dx = -e^{-x}(x^2 + 2x + 2)$ (by parts twice) · $\int \cos u\,du = \sin u$ · $\int \sin u\,du = -\cos u$. Evaluate $[F(x)]_a^b = F(b) - F(a)$: at $x = \infty$ only $e^{cx}$ with $c < 0$ becomes 0, at $x = 0$ every $x$ is 0 so only the plain number is left → $\int_0^\infty e^{-cx}dx = \frac{1}{c}\cdot\int_0^\infty xe^{-cx}dx = \frac{1}{c^2}\cdot\int_0^\infty x^2e^{-cx}dx = \frac{2}{c^3}$.

*Nov22-2f:* MMSE $= \dfrac{\int_0^\infty \theta\cdot\theta e^{-(x+a)\theta}d\theta}{\int_0^\infty \theta e^{-(x+a)\theta}d\theta} = \dfrac{2/(x+a)^3}{1/(x+a)^2} = \dfrac{2}{x+a}$.

$\ln$ **and** $e$**.** $\ln(ab) = \ln a + \ln b \cdot \ln\frac{a}{b} = \ln a - \ln b \cdot \ln a^k = k\ln a \cdot \ln e^a = a \cdot e^{\ln a} = a \cdot e^a e^b = e^{a+b}$.

Get the letter alone: $e^{ky} = A \to y = \frac{\ln A}{k} \cdot x^k = a \to x = a^{1/k} \cdot y = k\ln x \to x = e^{y/k}$.

**Derivatives.** $(c)' = 0 \cdot (\theta^k)' = k\theta^{k-1} \cdot (1/\theta)' = -1/\theta^2 \cdot (\ln\theta)' = 1/\theta \cdot (e^{u(\theta)})' = e^{u(\theta)}u'(\theta) \cdot [\ln f(\theta)]' = f'(\theta)/f(\theta) \cdot (uv)' = u'v + uv' \cdot \cos' = -\sin$, $\sin' = \cos$. Differentiating with respect to $\theta$ means all $x_n$ are constants.

**Sums.** $\sum_{n\geq 0} q^n = \frac{1}{1-q}$ for $|q| < 1$ · $\sum_n x_n = N\bar{x}$ · $\sum_n c = Nc$ · $\sum_n(x_n - \bar{x}) = 0$ · $\sum_n(ax_n + b) = a\sum_n x_n + Nb$.

**Fractions & equations.** $\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$ · $\frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$ · $\frac{a}{b} = \frac{c}{d} \to ad = bc$ · $ax^2 + bx + c = 0 \to x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ · multiplying by a negative flips $>$ into $<$ · $|x| > t =$ two cases.

**Numbers / Gaussian tail.** $\ln 2 = 0.693$ · $\ln 3 = 1.099$ · $\ln 10 = 2.303$ · $e^{-2} = 0.135$ · $\sqrt{2\pi} = 2.5066$ · $Q(-t) = 1 - Q(t)$ · $Q(0) = 0.5$ · $Q(1) = 0.1587$ · $Q(1.28) = 0.10$ · $Q(1.645) = 0.05$ · $Q(1.96) = 0.025$ · $Q(2.33) = 0.01$. For a two-sided interval $\pm t$, central area $= 1 - 2Q(t)$.

## Calculation rescue · when an algebra step stops you

**1 · Exponential integral: antiderivative, THEN limits.** To calculate $\int_0^\infty \theta e^{-b\theta}d\theta$ with $b > 0$, use the $xe^{cx}$ formula above with $x = \theta$, $c = -b$: $\int \theta e^{-b\theta}d\theta = -e^{-b\theta}(\theta/b + 1/b^2)$. At $\infty$, $e^{-b\theta} = 0$, so the upper value is 0. At 0, the value is $-1/b^2$. Upper minus lower gives $\boxed{0 - (-1/b^2) = 1/b^2}$. Do not forget to subtract the lower value.

**2 · Where the Beta MAP formula comes from.** If $q = \theta^A(1 - \theta)^B$, then $\ln q = A\ln\theta + B\ln(1 - \theta)$. Differentiate: $A/\theta - B/(1 - \theta)$. Set equal to 0: $A/\theta = B/(1 - \theta) \to A(1 - \theta) = B\theta \to A = (A + B)\theta \to \boxed{MAP = A/(A + B)}$. This is why you only need the final exponents in 2B.

**3 · Where the Gamma MAP formula comes from.** If $q = \lambda^A e^{-B\lambda}$, then $\ln q = A\ln\lambda - B\lambda$. Differentiate: $A/\lambda - B$. Set to $0 \to A/\lambda = B \to A = B\lambda \to \boxed{MAP = A/B}$. MMSE is NOT the peak; use the table's posterior mean $(A + 1)/B$.

**4 · A negative power in an MLE derivative.** $\frac{d}{d\theta}[-\frac{S}{2\theta}] = \frac{d}{d\theta}[-\frac{S}{2}\theta^{-1}] = -\frac{S}{2}(-1)\theta^{-2} = \boxed{+\dfrac{S}{2\theta^2}}$. Here $S = \sum x_n^2$ is constant while differentiating with respect to $\theta$. The two minus signs cancel.

**5 · Solving an exponential threshold.** $e^{-ct} = a \to$ take ln: $-ct = \ln a \to \boxed{t = -\ln(a)/c}$. Example: $e^{-2t} = 0.1 \to -2t = \ln 0.1 = -2.303 \to t = 1.151$. If the equation has $e^{-ct}$ on the OTHER side, first isolate it by division.

**6 · Quadratic formula and choosing the MA root.** For $b^2 + 8b + 1 = 0$, use $a = 1, b_{quad} = 8, c = 1$: $b = \frac{-8 \pm \sqrt{64 - 4}}{2} = -4 \pm \sqrt{15}$. The two values are $-0.127$ and $-7.873$; MA(1) keeps $\boxed{-4 + \sqrt{15} = -0.127}$ because $|b| < 1$. Do not confuse the coefficient $b$ in the quadratic formula with the MA parameter.

**7 · AR(2): the jump from coefficients to the final PSD.** Jun26 has $a_1 = -\frac{8}{5}$, $a_2 = \frac{1}{5}$, $\sigma^2 = \frac{8}{5}$. Compute the denominator pieces separately: $1 + a_1^2 + a_2^2 = 1 + \frac{16}{25} + \frac{1}{25} = \frac{42}{25}$ · $2a_1(1 + a_2) = 2(-\frac{8}{5})(\frac{6}{5}) = -\frac{96}{25}$ · $2a_2 = \frac{2}{5}$. Therefore $P = \dfrac{8/5}{42/25 - (48/25)\cos\theta + (2/5)\cos 2\theta}$. Now replace $\cos 2\theta = 2\cos^2\theta - 1$: constant $= \frac{42}{25} - \frac{2}{5} = \frac{32}{25}$ and $\cos^2$ coefficient $= \frac{48}{5}$. Multiplying top and bottom by 25 gives $\dfrac{40}{32 - 48\cos\theta + 20\cos^2\theta} = \boxed{\dfrac{10}{8 - 12\cos\theta + 5\cos^2\theta}}$.

**8 · Correlogram: the jump to a perfect square.** Let $c = \cos\theta$. For Jun26: $3 + 4c + 2\cos 2\theta = 3 + 4c + 2(2c^2 - 1) = 3 + 4c + 4c^2 - 2 = \boxed{4c^2 + 4c + 1 = (2c + 1)^2}$. A squared PSD is nonnegative, which is a useful check.

**9 · Sums with actual numbers.** For $x = [1, 2, 3]$, $N = 3$ and $\bar{x} = (1 + 2 + 3)/3 = 2$. $\sum x_n^2 = 1^2 + 2^2 + 3^2 = 14$ · $\sum(x_n - \bar{x})^2 = (-1)^2 + 0^2 + 1^2 = 2$ · $\hat{r}[0] = 14/3$ · $\hat{r}[1] = (1 \cdot 2 + 2 \cdot 3)/3 = 8/3$ · $\hat{r}[2] = (1 \cdot 3)/3 = 1$. The sums have fewer products as the lag grows, but the denominator stays $N = 3$.

**10 · $P_{FA}$ and $P_D$: same decision region, different pdf.** Suppose you decide $H_1$ when $x > t$, with $p(x \mid H_0) = e^{-x}$ and $p(x \mid H_1) = 2e^{-2x}$, $x \geq 0$. Then $P_{FA} = \int_t^\infty e^{-x}dx = [-e^{-x}]_t^\infty = 0 - (-e^{-t}) = e^{-t}$. Use the SAME limits under $H_1$: $P_D = \int_t^\infty 2e^{-2x}dx = [-e^{-2x}]_t^\infty = e^{-2t}$. If $P_{FA} = 0.1$, then $e^{-t} = 0.1 \to t = -\ln 0.1 = 2.303$ and $P_D = e^{-2(2.303)} = 0.01$. Do not integrate $H_1$ for $P_{FA}$.